

JIO SUBSCRIBER RETENTION & AI CHATBOT

Final Business Report

Capstone Project Submission

Project	Status
Dataset	Synthetic Jio retention sandbox dataset
Subscribers analysed	64,738
30-day churn rate	1.73%
90-day churn rate	5.01%
Final project validation	95 passed / 0 failed / 0 warnings
Implementation stack	Python, MySQL, CatBoost/XGBoost, SHAP, Ollama/Qwen, Streamlit

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1. Executive Summary

This project delivers an end-to-end subscriber-retention analytics solution using the supplied synthetic Jio sandbox dataset. The work moves from business diagnostics to SQL validation, 30-day churn modelling, customer value prioritization, offer-response feasibility analysis, and a privacy-safe employee analytics chatbot.

Management headline

The analysis identifies consistent risk patterns around recharge inactivity, early tenure, unresolved service issues, network congestion, and lower ARPU. The predictive workflow adds a subscriber-level risk score, while the risk-value layer separates high-risk customers by estimated customer value. The chatbot then exposes aggregate business intelligence through a controlled natural-language interface.

KPI	Result	Business interpretation
Subscribers	64,738	Synthetic sandbox base used for modelling
30-day churn	1.73%	Primary predictive target
90-day churn	5.01%	Secondary longer-horizon outcome
Top-decile test lift	6.78x	Risk model concentrates churn above the base rate
Test recall at top decile	67.86%	Share of test churners captured in highest-risk decile
Offer exposure	13,991 (21.61%)	Observed historical offer-exposed population
Offer redemption/response	37.12% of exposed	Descriptive historical response rate
Final validation checks	95 passed / 0 failed / 0 warnings	Project package and code integrity check

2. Business Problem and Objectives

The supplied brief frames retention as a diagnostic, predictive, prescriptive, and conversational-analytics problem. For diagnostics, it asks the team to establish where churn concentrates and which observable levers are associated with exits before building predictive models. The brief specifies a written findings note and an executive dashboard with circle, segment, and tenure drill-downs.

- Identify circles and customer segments with materially different churn rates.
- Understand churn patterns by tenure, ARPU, recharge behaviour, complaints, service requests, and network quality.
- Build a 30-day churn probability workflow using imbalance-aware evaluation and threshold optimization.
- Add customer value so retention activity is not driven by risk alone.
- Assess whether historical offer data supports response prediction or causal uplift.

- Build a controlled employee chatbot that converts business questions to safe SQL and produces traceable answers.

The project deliberately distinguishes descriptive association from causation. Differences in churn across segments identify useful signals, but they do not prove that changing one feature alone will prevent churn.

3. Data and Analytical Scope

Dataset	Rows	Role
subscribers	64,738	Customer-level churn, tenure, ARPU, recharge, network and complaint features
service_requests	14,206	Ticket category, status, SLA and resolution context
network_sites	3,847	Congestion, SINR, drop-call and throughput context
circle_monthly_kpi	396	18-month circle KPI history for trend analysis
circle_targets	22	Circle targets and review/budget metadata
offer_catalogue	20	Retention offer catalogue and expected uplift metadata

The primary target is churn_flag_30d. For honest modelling, future-outcome fields such as mnp_enquiry_flag, churn_reason, churn_date, and the 90-day target were excluded from the 30-day model feature matrix; subscriber_id was treated as an identifier rather than a predictive feature.

4. Diagnostic Business Findings

The diagnostic phase answers the core questions in the brief using the synthetic subscriber, service-request, network, and circle KPI data.

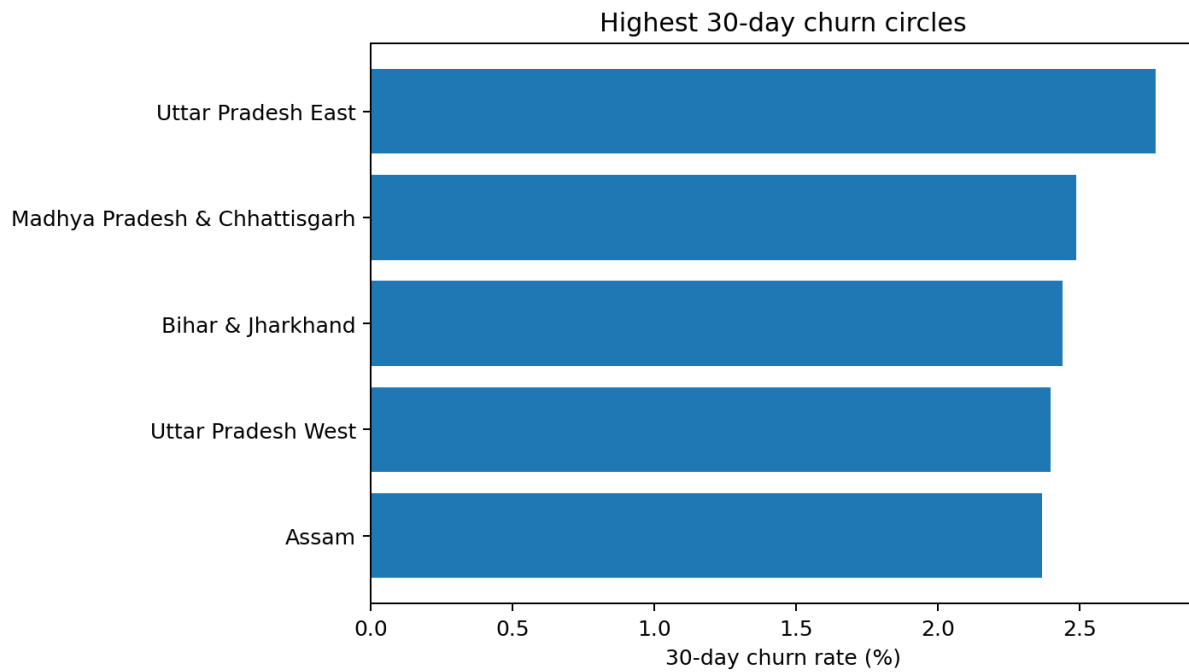


Figure 1. Highest observed 30-day churn rates by circle in the subscriber-level sandbox.

Circle-level churn is concentrated rather than uniform. The five highest observed rates are Uttar Pradesh East (2.77%), Madhya Pradesh & Chhattisgarh (2.49%), Bihar & Jharkhand (2.44%), Uttar Pradesh West (2.40%), and Assam (2.37%).

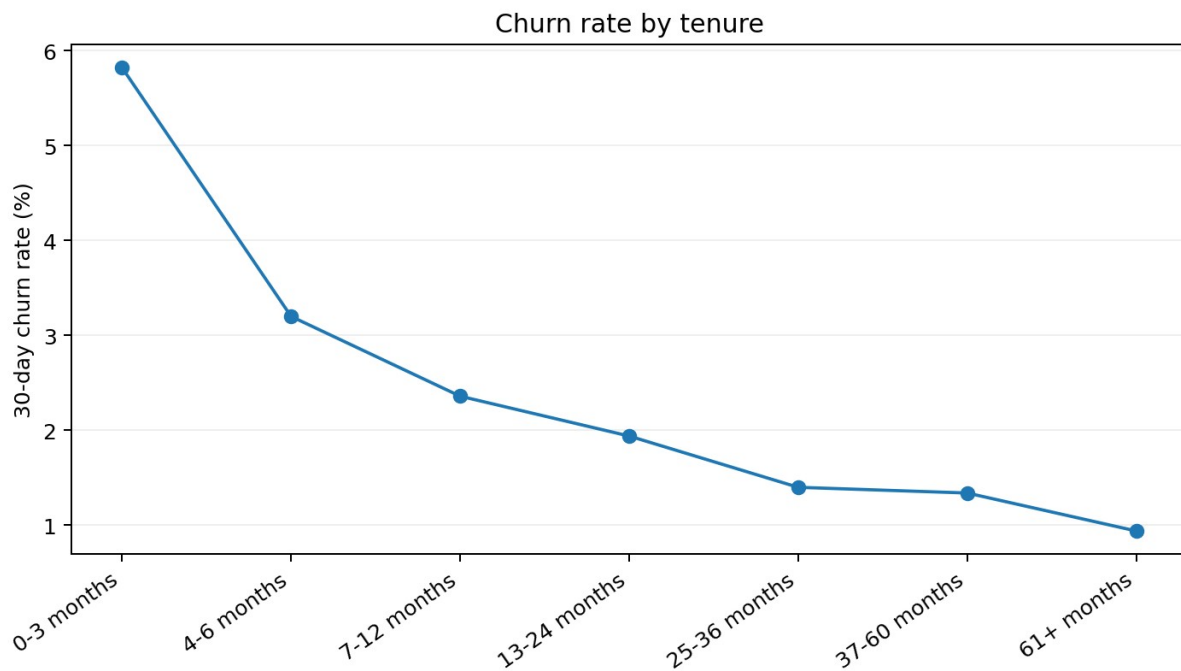


Figure 2. 30-day churn rate declines across the tenure bands in the supplied snapshot.

Tenure is strongly associated with churn in the sandbox: 0–3 month subscribers show 5.82% churn versus 0.94% for the 61+ month group. This makes the early customer lifecycle a clear monitoring segment for onboarding and first-90-day experience analysis.

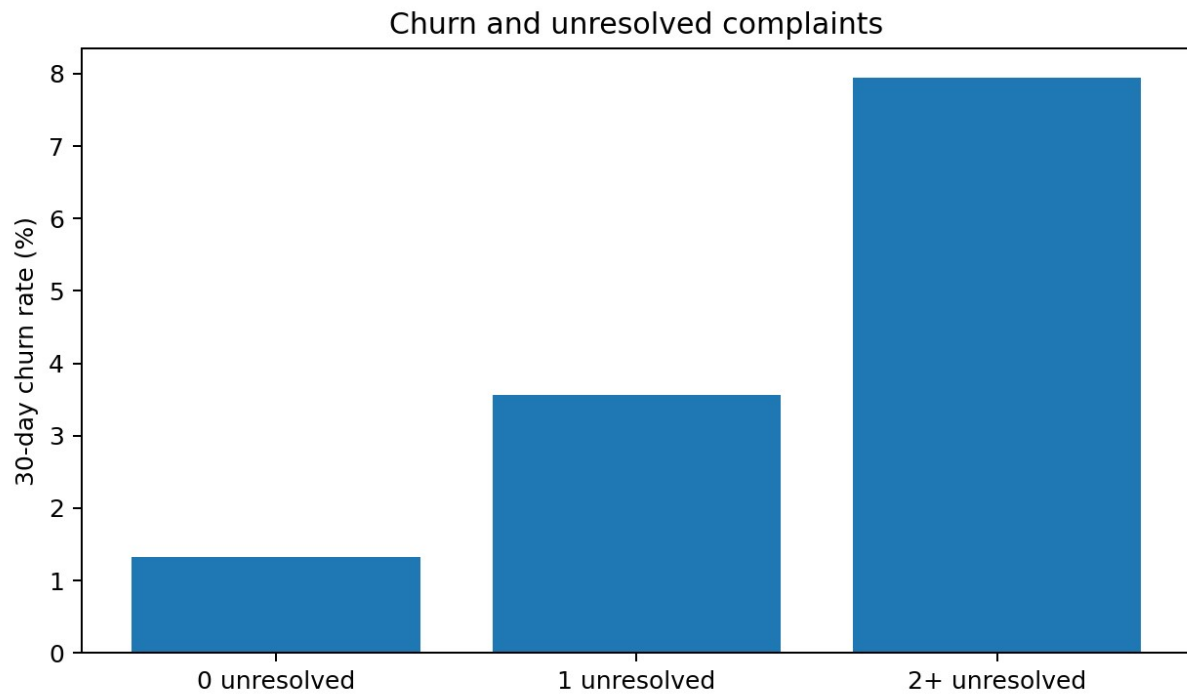


Figure 3. Observed churn rises with unresolved complaint burden.

Unresolved service issues show a large separation in the descriptive data. Churn is 1.32% with no unresolved complaints, 3.57% with one, and 7.95% with two or more.

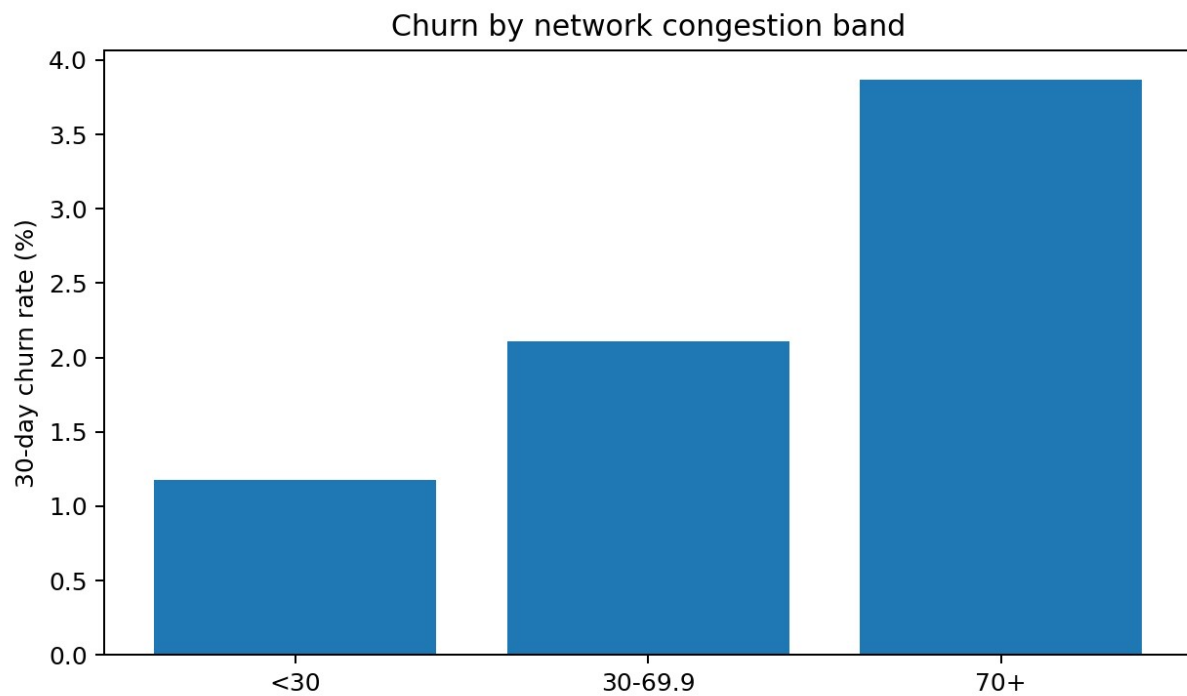


Figure 4. Observed churn rises with network congestion.

The congestion gradient is also pronounced: 1.18% churn below a congestion score of 30, 2.11% in the 30–69.9 range, and 3.87% at 70+.

Pattern	Observed result	Business implication
Recharge inactivity	61+ days since recharge: 6.09% churn	Use recency/gap measures as early-warning triggers
ARPU band	<₹150: 3.22%; ₹250+: 0.51%	Retention economics should account for value differences
Complaints	2+ complaints: 3.32% churn	Ticket volume is a useful service-friction signal
Service requests	0 requests: 1.61%; 2+: 3.22%	Repeated customer contacts merit monitoring
Network quality	SINR <5 dB: 2.56%; 15+ dB: 1.29%	Poor radio conditions correlate with higher churn

The strongest numeric correlations in the exploratory analysis were `days_since_last_recharge` and `avg_recharge_gap_days`, followed by recharge frequency, payment failures, unresolved complaints, app activity, ARPU, and congestion. These are associations, not causal estimates.

5. Data Quality and Governance

The data-quality stage performed 77 checks and identified one substantive review issue plus model-leakage items. No source rows were deleted and the supplied churn target was preserved.

Control	Result
Subscriber IDs	64,738 rows and 64,738 unique IDs
Service-request referential integrity	14,206 records matched valid subscribers; unmatched = 0
Synthetic target/date inconsistency	739 records have churn_flag_30d = 1 with churn_date beyond the 30-day horizon
Leakage controls	mnp_enquiry_flag, churn_reason, churn_date and future 90-day target excluded from 30-day model features
SQL validation	Six MySQL tables loaded and core Python findings reproduced with SQL
Chatbot access	Read-only jio_chatbot database user with approved-table allow-list

Governance decision

The supplied churn_flag_30d field is treated as the authoritative target despite the documented synthetic date inconsistency. churn_date is not used as a predictor.

6. 30-Day Churn Predictive Modelling

Because the 30-day churn event rate is only 1.73%, model evaluation emphasizes recall, precision, F1, ROC-AUC, PR-AUC, recall at the top decile, and lift rather than accuracy alone. The workflow used a stratified 60/20/20 train/validation/test split and kept the test set untouched until final evaluation.

Model / stage	Validation PR-AUC	Validation ROC-AUC	Recall @ Top-Decile	Top-Decile Lift
Logistic Regression baseline	0.2256	0.9204	75.56%	7.55x
Gradient Boosting	0.2249	0.9174	74.22%	7.42x
XGBoost original	0.2117	0.9158	72.44%	7.24x
CatBoost original	0.2357	0.9181	73.78%	7.38x
Regularized / tuned candidates	0.2206–0.2304	0.9170–0.9194	72.44–76.00%	7.24–7.60x

The final implementation used the original CatBoost model with a 0.49 operating threshold. This choice was carried forward to the untouched test set after validation and threshold analysis; it should be treated as the project implementation configuration rather than as a universal optimum.

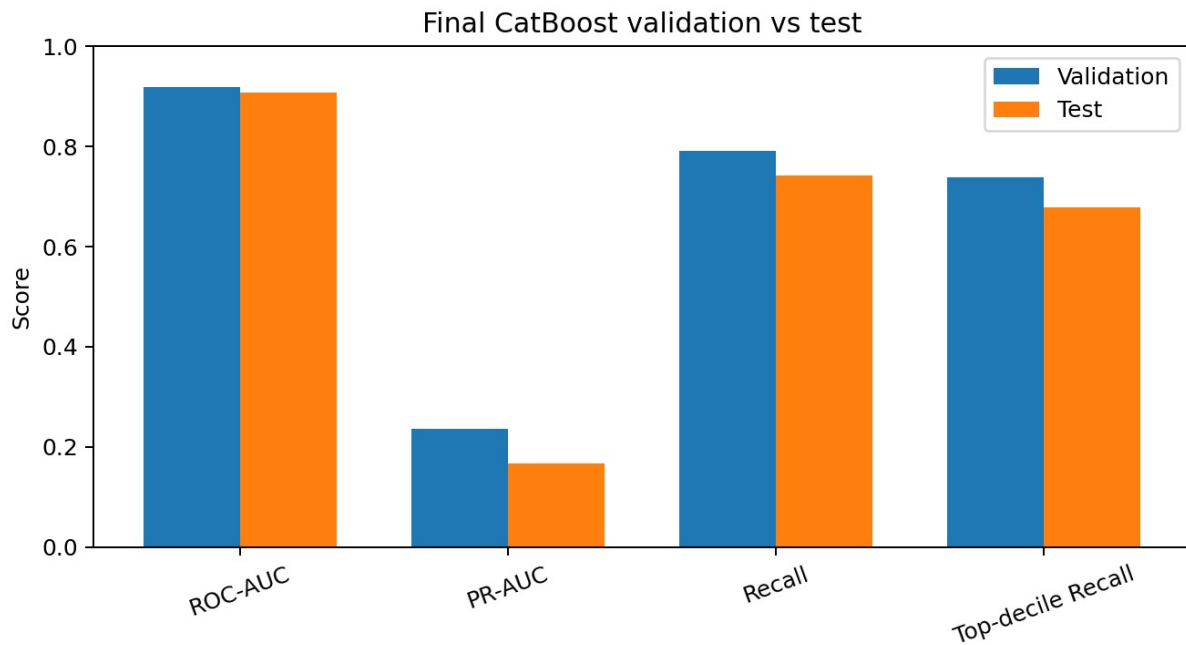


Figure 5. Final CatBoost validation and untouched test-set metrics.

Test metric	Final result
Accuracy	0.8845
Precision	0.1035
Recall	0.7411
F1	0.1816
ROC-AUC	0.9082
PR-AUC	0.1677
Top-decile Recall	0.6786
Top-decile Lift	6.78x

On the untouched test set, the final model achieved 0.9082 ROC-AUC, 0.1677 PR-AUC, 74.11% recall, 10.35% precision, F1 of 0.1816, 67.86% recall in the top decile, and 6.78x lift. The test result is lower than validation, which is expected and is explicitly retained as the out-of-sample performance view.

Feature signals

- days_since_last_recharge was the most prominent signal in the original CatBoost feature-importance output.
- tenure_months, outgoing_to_competitor_pct, avg_sinr_db, site_congestion_score, ARPU measures, avg_recharge_gap_days and app activity also ranked prominently.
- Feature importance indicates model contribution/association; it does not establish causal impact.

7. Customer Value and Risk Prioritization

A genuine future 12-month CLV target was not available in the sandbox. Instead of fabricating a supervised CLV label, the project uses a transparent value proxy based on annualized 6-month average ARPU. Risk and value are then combined into operational segments.

Segment	Test customers	Test churners	Observed 30-day churn
Priority 1: High Risk / High Value	101	13	12.87%
Priority 2: High Risk / Upper-Mid Value	250	28	11.20%
Priority 3: High Risk / Lower Value	943	111	11.77%
Priority 4: High Value Watchlist	223	3	1.35%
Standard Monitoring	11,431	69	0.60%

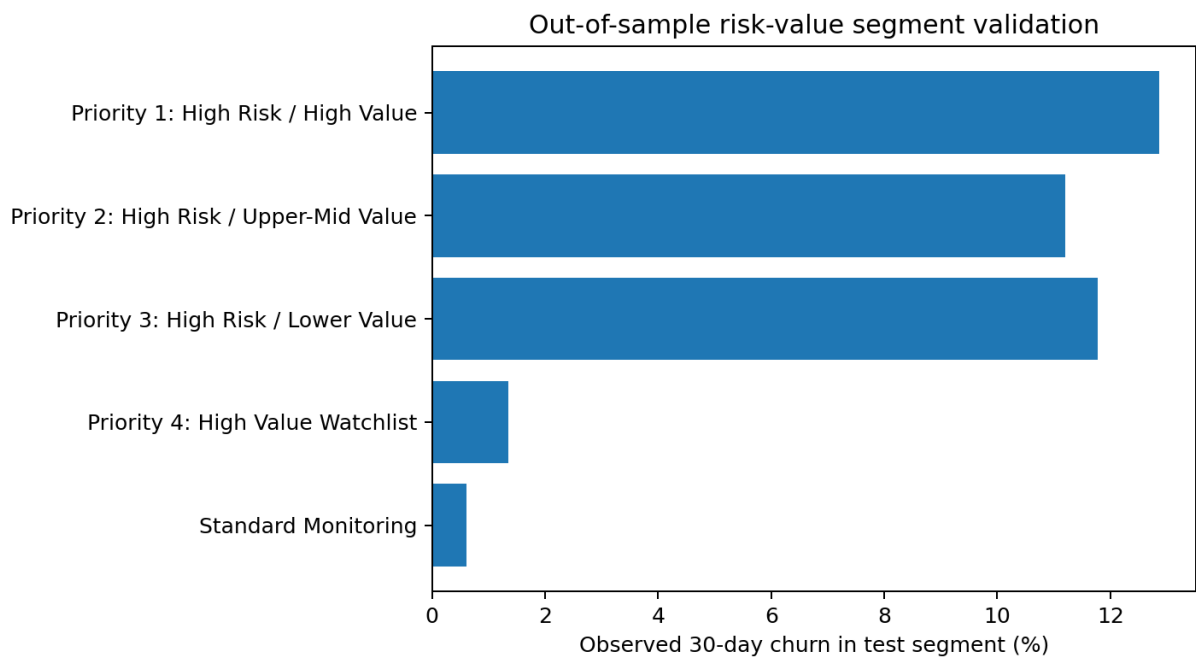


Figure 6. Out-of-sample churn rates for risk-value segments.

The test set preserves clear separation between high-risk value segments and the standard-monitoring population. Priority 1 contains 101 test customers with 13 observed churners (12.87%), compared with 0.60% for standard monitoring. This supports the use of risk-value segmentation as an operational layer on top of the churn score.

8. Offer Response and Uplift Feasibility

Historical offer exposure covers 13,991 subscribers, or 21.61% of the subscriber base. Among exposed subscribers, 5,194 redeemed the offer, producing a 37.12% historical response rate.

Population	Count	30-day churn rate
Not exposed	50,747	1.70%
Exposed	13,991	1.84%
Exposed / not redeemed	8,797	1.82%
Exposed / redeemed	5,194	1.89%

The observed churn differences are descriptive only because the available data does not provide a randomized treatment/control design or a reliable subscriber-offer treatment assignment structure. Therefore, causal uplift claims are not made.

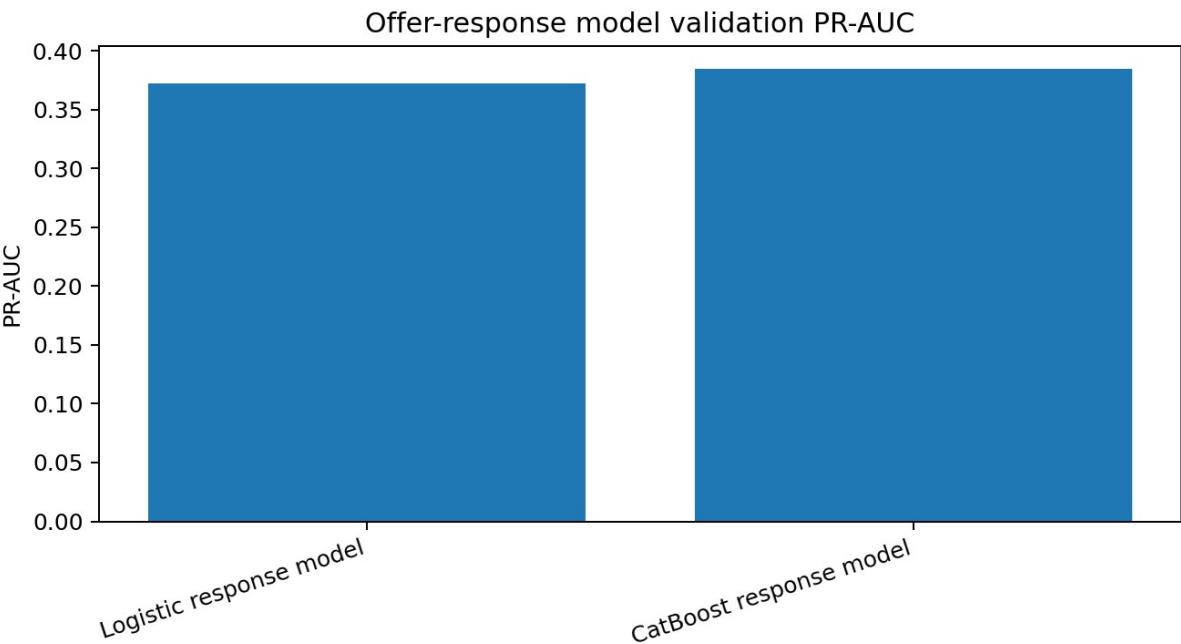


Figure 7. Validation PR-AUC for the offer-response models; the observed signal is close to the no-skill response baseline.

Model	Validation PR-AUC	Top-10% lift
Logistic response model	0.3725	1.04x
CatBoost response model	0.3847	1.11x

The response branch was closed as a feasibility finding rather than promoted to a retention decision model. A controlled experiment with treatment/control assignment would be required before estimating incremental uplift or treatment effect.

9. AI Employee Chatbot

The chatbot implements the conversational analytics layer described in the architecture: an employee asks an English business question, a local Qwen 2.5 3B model generates a SQL candidate, the SQL passes the read-only security engine, MySQL executes it, and the returned data is summarized with verified Python logic for key numeric answers.

Architecture

Streamlit UI -> privacy filter -> local Qwen 2.5 3B -> SQL generation -> Step 31 SQL validation -> read-only MySQL -> sanitized result -> verified answer synthesis -> audit log.

Security controls

- Natural-language privacy filtering blocks requests for subscriber identifiers and employee personal/sensitive information.
- Approved-table allow-list controls which database tables the SQL agent can query.
- SELECT-only validation blocks DELETE, UPDATE, DROP, and other modifying statements.
- Sensitive-column blocking prevents fields such as subscriber identifiers from being returned.
- Automatic row limits and result sanitization reduce accidental over-disclosure.
- The database connection uses the read-only jio_chatbot MySQL user.
- Timestamped audit logging captures question, generated SQL, validation/execution status, runtime, row count and result/error information.

SHAP explainability

The predictive workflow now includes SHAP explainability on the existing final CatBoost churn model. Global mean-absolute SHAP values identify influential model features, while privacy-safe individual reason codes support high-risk case explanations. Subscriber identifiers are retained only for internal matching and are excluded from user-facing SHAP explanations. SHAP is used as model explanation and does not establish causal relationships.

Demonstrated UI behaviour

Scenario	Observed behaviour
Business question: top 5 circles by 30-day churn	SQL generated, validated, executed, result returned and summarized
Business question: average ARPU by circle	SQL result verified in Python; highest = Himachal Pradesh ₹219.46 and lowest = North East ₹209.68
Sensitive request: all subscriber IDs	Blocked by privacy filter before execution
Modification request: delete subscribers with zero complaints	Blocked by SELECT-only SQL security layer

10. Business Implications and Operating Model

The completed workflow converts descriptive risk signals into operationally usable retention intelligence. The following actions are directly linked to the project evidence and architecture rather than being presented as causal claims about customer behaviour.

1. Create an early-life monitoring view for 0–3 month subscribers and track onboarding/service friction during the first 90 days.
2. Use unresolved-service thresholds as operational triggers for escalation and service recovery analysis.
3. Combine recharge recency/gap behaviour with churn risk so silent disengagement can be detected before a formal port-out signal.
4. Overlay network congestion and SINR context when reviewing high-risk subscriber clusters or circles.
5. Use risk × value segments to distinguish high-risk/high-value populations from high-risk/lower-value populations for different retention workflows and economics.
6. Keep offer intervention evaluation experimental: establish treatment/control assignment before making incremental uplift claims.
7. Use the chatbot as a governed self-service analytics interface, with the current security controls retained as non-negotiable guardrails.

Suggested phased operating model

Phase	Scope	Current project status
1. Diagnostics	Circle, segment, tenure, complaint, network, recharge and ARPU analysis	Completed
2. Predictive retention	30-day churn risk + value prioritization	Completed for capstone
3. Offer experimentation	Controlled treatment/control design for uplift	Feasibility established; experiment required
4. Conversational analytics	Employee SQL assistant + governed answer synthesis	Working Streamlit prototype with governed SQL and SHAP explanations
5. Scale-up	Production integrations, monitoring, model governance and retrieval expansion	Future work

11. Limitations and Next Steps

- The dataset is synthetic and should not be interpreted as a live Jio customer extract.
- The 30-day target contains a documented synthetic churn-date inconsistency affecting 739 records; the target flag was preserved intentionally.
- CLV is a transparent ARPU-based value proxy, not a supervised forecast of actual 12-month contribution.

- Historical offer data is not sufficient for causal uplift estimation; randomized or defensible treatment/control assignment is needed.
- The current chatbot reads structured MySQL data. The architecture brief also calls for controlled document retrieval over offers, targets and price sheets; that retrieval layer is not yet part of the working prototype.
- The final chatbot uses a local Qwen 2.5 3B model. A production implementation would require a stronger evaluation set, monitoring, model/version governance and infrastructure hardening.
- The architecture brief also calls for a certified golden question-answer evaluation set; that remains a natural production-hardening enhancement beyond the current capstone prototype.

12. Conclusion

The Jio Subscriber Retention & AI Chatbot capstone now forms a complete business-analytics workflow: the diagnostic stage identifies where churn is concentrated and which observable customer/network signals are associated with it; the modelling stage produces an imbalance-aware churn risk score; the value layer adds economic context; the offer-response analysis establishes the limits of historical data for causal uplift; and the chatbot exposes validated, privacy-safe business intelligence through a local LLM and read-only SQL path.

The completed package was subjected to final project validation with 95 checks passed, 0 failed and 0 warnings. SHAP explainability has been added to the churn-model workflow and connected to the Streamlit chatbot for privacy-safe model-factor explanations. The GitHub repository is maintained as a private repository, consistent with the confidentiality marking in the supplied project brief.

Appendix A. Key Project Artifacts

Area	Primary artifacts
Diagnostics	outputs/reports/04_business_analysis_report.txt; outputs/charts/01–11
Modeling	outputs/reports/15–20; models/*.pkl
CLV / value	data/clv/clv_value_prepared.csv; data/clv/customer_risk_value_priority.csv
Offer response	outputs/reports/24–28
Chatbot	src/29-35, 38; data/modeling/chatbot_safe_schema.json; outputs/reports/33-34; SHAP artifacts in outputs/reports/37* and outputs/charts/25_shap_global_feature_importanc e.png
Validation	src/36_final_project_validation.py; outputs/reports/36_final_project_validation_report .txt
Architecture	architecture/Jio_Architecture_Plan.txt; architecture/Jio_End_to_End_Architecture.pdf/png

Submission support	README.md; requirements.txt; .env.example; .gitignore
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Appendix B. Suggested Live Demo Flow

1. Show the private GitHub repository and README.
2. Open the architecture diagram and explain the end-to-end data/AI flow.
3. Present the main diagnostic findings: circle, tenure, complaints, network, ARPU and recharge patterns.
4. Present the final churn test results and top-decile lift.
5. Show the risk $\times$ value segment validation.
6. Explain the offer-response feasibility result and why causal uplift was not claimed.
7. Present the SHAP explainability output and explain that it provides model-based, non-causal reason codes.
8. Start Streamlit with ``python -m streamlit run src/35_jio_chatbot_app.py``.
9. Ask: “Which 5 circles have the highest 30-day churn rate?” and show the validated SQL/result.
10. Ask: “What is the average ARPU by circle?” and show the verified highest/lowest values.
11. Test “Give me all subscriber IDs.” and “Delete all subscribers with zero complaints.” to demonstrate privacy and read-only SQL controls.

Source and Method Note

This report is based on the supplied Jio_Retention_Brief.pdf, the synthetic Jio_Retention_Dataset.xlsx and the completed project artifacts/results generated during the capstone workflow. Where the source brief provides business context or directional external estimates, they are treated as context rather than project-measured results. All customer-level analytical results in this report are from the supplied synthetic sandbox data.